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# Simulation results

*massive-binary-sim v0.1.0 · git `UNKNOWN(no` · seed 42 · wallclock 0.04 s*

## 1. Initial conditions

| quantity | value |
|---|---|
| M1, M2 (ZAMS) | 30, 12 Msun |
| separation / period | 45 Rsun / 5.396 d |
| eccentricity | 0 |
| metallicity Z | 0.0142 |
| spins (Omega/Omega_crit) | 0, 0 |

### Active prescriptions

| toggle | choice |
|---|---|
| gravity | `2.5PN` |
| wind | `vink2001` |
| convection | `ledoux` |
| overshoot | `exponential` |
| am_transport | `taylor_spruit` |
| mt_stability | `zeta` |
| mt_rate | `kolb_ritter` |
| ce_formalism | `alpha_lambda` |
| opacity | `kramers` |
| reaction_rates | `reaclib` |
| sn_engine | `delayed` |
| tides | `hut1981` |
| structure_fidelity | `semi_analytic` |
| third_body | `False` |

## 2. Evolutionary timeline

| t [Myr] | event | detail |
|---|---|---|
| 0.019 | star 1 -> MS | reduced-model phase |
| 6.506 | star 1 -> HG | reduced-model phase |
| 6.631 | star 1 -> CHeB | reduced-model phase |
| 0.019 | star 2 -> MS | reduced-model phase |
| 6.631 | **common_envelope** | merger: a_f=9.509 Rsun < 1.5(R_core+R_comp) |

**Fate:** common-envelope merger (Case B, q=2.42)

## 3. Final-state table

| quantity | star 1 | star 2 |
|---|---|---|
| mass [Msun] | 28.98 | 11.96 |
| phase | CHeB | MS |
| separation | 45.56 Rsun | |
| period | 5.567 d | |
| eccentricity | 0 | |

## 4. Derived quantities

| quantity | value | unit | formula |
|---|---|---|---|
| chirp mass M_c | 15.9 | Msun |  $M_c = (m_1 m_2)^{3/5} / (m_1+m_2)^{1/5}$  |
| mass ratio q | 0.4126 | - |  $q = m_2 / m_1 (<=1)$  |
| symmetric mass ratio eta | 0.2068 | - |  $\eta = m_1 m_2 / (m_1+m_2)^2$  |

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| integrated wind loss (star1 / star2) | 1.012 | Msun | `int |Mdot_wind,1| dt ; star2 = 0.0414 Msun` |
| integrated RLOF transfer | 0 | Msun | `int |Mdot_RLOF| dt (gross, donor side)` |
| net Delta M_1 | 1.012 | Msun | `M_1(ZAMS) - M_1(final)` |
| net Delta M_2 | -0.04151 | Msun | `M_2(final) - M_2(ZAMS)` |
| orbital separation (final) | 45.56 | Rsun | `state variable` |
| orbital period (final) | 5.567 | day | `P = 2 pi sqrt(a^3 / G(m1+m2))` |
| orbital binding energy | -1.4428e+49 | erg | `E = -G m1 m2 / (2a)` |
| orbital angular momentum | 2.2092e+54 | g cm^2 / s | `J = mu sqrt(G M a (1-e^2))` |
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5. Error budget

****Numerical error**** (from the convergence study):

- not run in this invocation (`--convergence` flag)

****Physical / model uncertainty**** dominates and is quantified in §6-7. The reduced single-star evolution model (fitting formulae, not a structure)

6. Free-parameter sensitivity (one-at-a-time)

not run (`--sensitivity`)

7. Theory comparison (competing prescriptions)

not run (`--theory-compare`)

8. Validation suite results

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| test | status | detail | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kepler_closure | PASS | 200 orbits: max|dE/E|=1.77e-11, max|dL/L|=8.76e-12, |da/a|=1.52e-11 (tol 1e-06) |
| lane_emen_analytic | PASS | n=0: max|dtheta|=5.72e-16, |dxil|=8.88e-16; n=1: max|dtheta|=1.55e-14, |dxil|=1.82e-14; n=5: max|dtheta|=5.44e-14, |dxil|=1.82e-14; n=10: max|dtheta|=5.44e-14, |dxil|=1.82e-14 |
| periastron_advance_1PN | PASS | 1PN d(omega)/orbit measured=6.157175e-05 rad, predicted 6 pi GM/(c^2 a(1-e^2))=6.155112e-05, frac diff=0.034% |
| hulse_taylor_decay | PASS | dPb/dt(Peters)=-2.402102e-12 s/s vs GR prediction -2.402630e-12 (Weisberg & Huang 2016); residual 0.022%. Merger |
| energy_conservation_2PN | PASS | 2PN conservative: secular |dE/E| over 120 orbits = 1.68e-08 (tol 1e-07); max|dL/L|=1.12e-05 |
| eddington_limit_flag | PASS | 120 Msun ZAMS Gamma Edd=0.406; 10x-luminosity state flags Gamma>1: True |
| solar_structure_eddington | PASS | Eddington model 1 Msun: P_c=1.256e+17 (ref 2.4e17), T_c=1.185e+07 (ref 1.57e7), rho_c=76.9 (ref 150), L=0.1 (ref 150) |
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9. Assumptions & limitations

- ****1-D, orbit-averaged.**** No 3-D hydrodynamics; the orbit is treated by secular equations except in the dedicated Kepler-closure test.
- ****Reduced stellar evolution.**** Single-star tracks are fitting formulae (Hurley+ 2000-style), not Henyey structure integrations. Core masses, radii and lifetimes carry ~10-30% systematic uncertainty.
- ****No magnetic fields**** beyond the Tayler-Spruit AM-transport toggle.
- ****No rotational mixing feedback**** on the tracks (rotation only sets Ω_{crit} and tidal rates here).
- ****Mass-transfer rate**** uses a Ritter/Kolb-Ritter-style analytic nozzle with an efficiency prefactor (KNOWN_GAPS BIN-MDOT).
- ****GW waveform**** is a 0PN Newtonian chirp -- qualitative, not for matched filtering.
- ****3.5PN dynamics**** not implemented (KNOWN_GAPS PN-35).
- Full list: KNOWN_GAPS.md.

11. Bibliography

See `REFERENCES.bib` for the full list with DOIs. Key prescriptions used in this run: Peters (1964); Mora & Will (2004); Eggleton (1983); Ritter